CAMPUSIM

An Integrated Parametric BIM for Campus Design Simulation and Optimization

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Abstract. Although simulation models have been recently employed to model and examine pedestrian behavior in urban areas, comparable research has not been pursued in campus environments despite their importance as a critical area of inquiry. Those models' paucity and methodological limitations suggest investigating new research and design strategies to objectively assess and describe how the qualities of campus spaces and zones influence human behavior and, hence, predict the patterns of users' interaction and space usage. Those patterns and their impact on health have been pointed out as critical to the relationship among public space and quality of life due to Covid-19. There is an urgent need to develop decision support tools that would support interactive design processes and enhance the quality of open space design in terms of sense of space, place-making, and user interaction. To that goal, this study has proposed the integrated parametric BIM-based campus life simulation "CampuSIM" as a method for parametrization of the qualities of pedestrian campus zones and spaces. The study proposed the use of multi-objective optimization methods to fulfill various campus quantifiable and non-quantifiable design objectives. The significance of the proposed tool will result from its potential application in a wide range of complex, dynamic pedestrian behavior scenarios such as flows, social simulations, and design.

Keywords. Campus Modelling, Campus Master Planning, Campus Design, Parametric Modelling, BIM, Design Optimization

1. Introduction

Empirical studies have shown that the quality and perceptions of the campus environment may affect the productivity of students, faculty, and staff; and students' retention, choice of higher education institution, and experience of academic life (Hajrasouliha 2017; Speake *et al.,* 2013). The users' perception of the campus's defining characteristics and outdoor spaces is closely tied to human activities (Deasy,

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1974; Duvon, 1966; Manning and Coleman-Boatwright, 1991) and will, over time, become tailored to the unique characteristics of the particular environmental setting where they habitually function (Abu-Ghazzeh, 1999). In recent years, most universities have reembraced the idea of campus master plans to address their institutional objectives, such as attracting more students, increasing the quality of life of current students and faculty, promoting a learning and research environment, creating a sustainable environment, and benefiting the surrounding communities (Hajrasouliha, 2017). The campus can thus provide a natural laboratory for examining the impact of the built environment characteristics on students' movement patterns, transportation, and walking behaviors. There is a paucity of research, however, about how individuals use and dynamically move between various campus zones despite its importance as a critical area of inquiry (Oppio *et al.*, 2021, Speake *et al.*, 2013; Abu-Ghazzeh, 1999).

2. Campuses as Pedestrian Districts

2.1. THE EFFECTS OF CAMPUS CHARACHTERISTICS ON USERS' BEHAVIOURS AND PERCEPCTIONS

Because a unique set of behaviours and perceptions can be associated with campuses, this study has focused on reviewing the studies that examined the characteristics of those environments and to understand the mechanism of choice on campus spaces.

There are four reasons why it is imperative to examine the relationship between the campus characteristics and students' behaviours in and interaction with campus spaces. First, a greater understanding of the decisional factors associated with active transportation can help universities, health promoters, and city planners design interventions to reduce barriers and encourage physical activity (Adkins *et al.,* 2012). Second, understanding such relationships from the perspective of a university planner could prove useful, as most colleges and universities in the past failed to emphasize the proven impact of the appearance of the physical environment on learning. Third, environmental perception helps campus users manage their communication and social interaction with each other, identify essential features of their everyday environment and enjoy aesthetic experiences (Koltsova *et al.,* 2012 a). Finally, the understanding of this relation becomes integral to the spatial design and decision-making of the campus environment (Zacharias, 2001).

There appear to be three main gaps in knowledge in this field. First, the identification of environmental factors in collective decision-making and their treatment in quasi-experiments tends the collected research toward cataloguing of effects rather than an explanation of underlying causes of such preferences (Ozkan, Alpak, and Var, 2017). Second, the high degree of subjectivity involved, and it follows that aesthetic issues are currently under-researched in urban planning in general and in transport planning in particular (Oppio *et al.,* 2021, Timms and Tight, 2010). Third, some studies have focused on micro-scale design elements but not as much on macro-scale design elements related to the contextual condition of the campus environment (Hajrasouliha, 2017). Hence, much uncertainty remains about the impact of the design features and urban layout of university campuses on the users and has been a major flaw in their planning and design (Alnusairat, Ayyad, and Al-Shatnawi, 2021).

The limitations of prior studies suggest using computer-performed simulations with new research and design strategies to dynamically investigate and empirically assess how those characteristics influence transportation-related activities and users' perceptions of quality to help identify designs and visual content that stimulate certain choices (Oppio *et al*., 2021, Zhou, 2008; Osaragi, 2004; Jalalian *et al.,* 2011; Bonabeau, 2002). This is particularly important for the heterogeneous campus population whose interactions are complex, nonlinear, discontinuous, or discrete.

2.2. MODELING TOOLS FOR SIMULATING PEDESTRIANS' BEHAVIOR

A variety of urban simulation models have been recently employed in architectural and urban design as decision-support tools to create an environment supporting interactive design processes such as accessing and visualizing urban design variables and generating a variety of design scenarios at city-wide and regional scales (Kolstova *et al*., 2012a). These models represent reliable tools for predicting and understanding the use of campus spaces and ultimately offer design decision-support that addresses questions such as: How well do designers predict the use of campus spaces they have created? How confident can they be that a place designed for certain types of activity and occupation will serve its users' needs well? (Golicnic and Thompson, 2010).

Agent-based simulation offers designers a powerful tool to model and simulate pedestrian behaviour at various scales (Ronald *et al.,* 2007). Because it helps identify, understand, and predict the complicated patterns of individual behaviour and assesses the outcomes of different scenarios and layouts of urban spaces (Zhou, 2008), its utility in campus design is one of its most powerful yet under-utilized applications. Advanced game engines provide real-time, interactive visualization that can be utilized for simulation and visualization of campus life and can use first or third-person perspectives. Various game engines have been used to visualize architectural design or human-environment interaction with a variety of software such as Torque Game Engine, VirTools®, Quest3DTM, CryENGINE®, Unity, and Unreal.

Parametric modelling represents a paradigmatic shift in building modelling methods from generating one solution to generating a system that produces multiple solutions used for optimization studies (Stocking, 2009; Eastman *et al.*, 2011). It was employed for campus design in a few cases in academia (Zuelzke *et al.,* 2012) but in very few cases in professional practice (Bullivant, 2012; Verebes, 2014).

To effectively use parametric modelling in campus master plans, designers should consider BIM's semantically rich and object-based features that can facilitate the creation and management of building data, which can be used in the design, simulation, construction, operation, and design optimization.

3. CampuSIM: Using Parametric Modelling for Design Optimization

3.1. DEVELOPMENT FROM OUR PREVIOUS WORK

The development of CampuSIM is a continuation of our previous work. The first work was a simulation system comprising an agent-based virtual user model and a built environment model representing geometric information and design elements' usability properties (Yan, 2014; Yan & Kalay, 2005). By inserting the virtual users in the built

environment model and letting them "explore" it, the system revealed the interrelationship between the environment and its users. The environment can then be modified to see how different planning affects user behaviours. Also, the cognitive modelling method that we used before could be extended for the proposed CampuSIM, which contains much more complex and comprehensive cognitive requirements.

We have also developed the environment-behaviour simulation with a new design process and supporting Building Information Modelling (BIM) and game tools (BIM-Game) (Yan, 2008; Yan, Culp & Graf, 2011). The project provided a framework for a real-time, interactive walkthrough of a Virtual User model in the interior environment. In the proposed CampuSIM, appropriate game engines will be investigated for simulating and visualizing the campus activities in both the exterior and interior spaces. In addition, we proposed a streamlined modelling process for the built environments: a new integrated parametric modelling, BIM, and building energy performance modelling process, which can provide easy-to-use and accurate simulation at the early design stage (Yan & Jeong, 2016) that enables design optimization.

This has inspired our proposed model: parametric BIM-based campus life simulation. By varying parameters of campus design, we can test each design option's performance in terms of users' behaviour and building's physical performance. Using parametric studies of the design options, we may achieve the optimization of campus design that will facilitate the better academic performance of students. In the proposed CampuSIM, an expanded streamlined modelling process that includes the Virtual User model will be applied to study the relationship between users' behaviour and the campus environment. Also, the functional parameters relating to campus users' behaviours could be embedded into the proposed system.

3.2. METHODOLOGY

The methodology comprises nonlinear approaches to the proposed model. The steps are i) defining measurable parameters/campus characteristics in the campus design context which impact pedestrian movements, ii) running pedestrian simulations in different scenarios based on destinations, iii) saving the people pass counts in specific locations in density maps, and iv) evaluating the generated data to cross-check with the actual scenario in the context. As a case site, we selected a state university campus in the United States and developed a primary 3D mass to feed into the simulation system.

The mixed-method approach employed combines quantitative BIM parametric modelling and observations through qualitative behaviour mapping to reveal common patterns of behaviour that appear to be correlated with particular layouts and details. The data from every observation, originally mapped by hand, will be recoded in digital form onto a GIS map system using ArcView. Conventional observation techniques offer a reliable tool for understanding environmental behaviour relationships. They would create rich, spatially explicit empirical databases of environment–behaviour interactions linked to spatial patterns (Koltsova *et al.* 2012a, b). These observations could be explored in several different ways according to the type of activities, estimated age of people observed, and weather conditions. Such mapped data allows analysis of the different spatial attributes of campus space use according to different data layers.

This study will examine and relate the following characteristics of campus spaces to the user's behaviours: a) urban layout (accessibility, spatial organizations, views); b)

physical features (seating, shading and rain shelter, surrounding buildings and facilities, landscape, greenery, areas for groups; c) design elements (colour, texture); d) outdoor thermal perceptions (air temperature/speed, solar radiation, relative humidity); and e) students' needs and behaviour (socializing, eating and drinking, relaxing, attending activities, exercising/walking, studying) (Alnusairat, Ayyad, and Al-Shatnawi, 2021).

3.3. GOAL: DESIGN INTEGRATION

The process of developing campus masterplans involves multiple stakeholders and multiphase developments plans. To make informed decisions, planners must formulate multiple development scenarios in a structured manner considering a variety of social, environmental, sustainable, and aesthetic factors. It is essential to address the challenge of computationally unifying the design and analysis cycles. Hence, this model is intended to integrate scenario generation with design automation and integration. The goal is to support designers in developing more accurate and detailed design alternatives that involve space exploration and analysis and ultimately to increase design space exploration and optimization (Kim *et al* 2015, Birch *et al*. 2013).

The process of constructing CampuSIM has considered the characteristics of the campus physical environment and pedestrian behaviour. Since the environment is a hybrid space, a mixture of large-scale and small-scale enclosed space, mixed-mode, and open space (Oppio *et al.,* 2021, Ronald *et al.*, 2007), it recommends creating more than one model to retrieve the required outputs.

3.4. UTILIZING BIM AND PARAMETRIC BIM IN CAMPUS MODELING

BIM is a process to facilitate the exchange and interoperability of building information (Eastman *et al.*, 2011) and a product of the physical and functional characteristics of a building. Its semantically rich and object-based features facilitate building data creation and management used in design, simulation, construction, and operation (Yan, 2014).

A key aspect of a BIM project is the concept that objects are defined by parameters and external data such as physical, aesthetic, and functional data. Parametric modelling enables these parameters to be processed by mathematical formulas and computational algorithms to update the model for exploring the search space of design solutions. Another key aspect is the concept of translating a data model into a working method, by which the model and its associated metadata become the primary repository of design information and tool for design communication (Meagher, 2015). Hence, its extensive impact on design practice increases the projects' efficiency and redefines how design knowledge gets shared across project stakeholders (Holzer, 2015).

In a previous study, we proposed a streamlined modelling process: a new integrated parametric modelling, BIM, and building energy performance modelling, which can provide accurate simulation at the early design stage when design strategies have the major influence on the building performance. An expanded streamlined modelling process that includes the Virtual User model will be applied to study the relationship between users' behaviour and the campus environment (Yan and Jeong, 2016).

3.5. MULTI-OBJECTIVE OPTIMIZATION

Like a building design, campus design can be seen as a Multi-Objective Optimization (MOO) process, aiming to find optimal solutions for multiple conflicting objectives. This is essential since multiple objectives will be used to evaluate the design's success. Unlike the energy performance and costs in our previous work, where the objectives (performance of buildings) are quantifiable, campus performance in terms of user behaviours may not be easily quantified and inserted into the optimization fitness functions. Therefore, multiple quantifiable and non-quantifiable objectives will be considered in optimizing a campus design, and creative design decisions for non-quantifiable objectives should be utilized (Yan *et al.*, 2015).

There are two methods to solve MOO problems (Su and Yan, 2015). The first is converting the problem into a single objective one by using a weighted sum of the objective functions with often arbitrary weights. The second is searching for the Pareto Optimal set, which is more widely accepted by MOO practitioners, as normally, they prefer to have access to multiple optimal design solutions. instead of a single solution. A change to a design option's parameter values that makes at least one objective better off without making any other objectives worse is called a Pareto improvement. A design option is defined as "Pareto Optimal" when no further Pareto Improvements can be made (Radford and Gero, 1987). For example, there can be multiple design options, each of which meets the following condition: any improvement of one objective such as decreasing the use of operating energy, will degrade at least one other objective such as increasing the life-cycle cost of the building. These design options are the Pareto Optimal Set we seek to find using computational methods.

The complexity of the model can increase when considering that users' behaviour can also affect quantifiable objective measures, for example, in energy consumption of buildings. Overall, for optimizing campus design, we needed an integrated parametric BIM and agent-based user model, which supports multi-objective design optimization through constructing fitness functions that will embed building performance simulations and user behaviours' simulations as part of the functions.

Existing BIM-based MOO tools such as Generative Design in Autodesk Revit or Rhino. Inside with the Grasshopper visual programming tool that run in Revit can be utilized to implement MOO in CampuSIM. CampuSIM will provide the design objectives and input parameters for the MOO tools to optimize through algorithms such as Genetic Algorithm or more specifically, NSGA-II, which is a fast elitist non-dominated sorting genetic algorithm (Deb et al., 2000).

3.6. CAMPUSIM: AN INTEGRATED MODEL

Campus design requirements and performance objectives come from planning and design programming. Design parameter identification and analysis require identifying design parameters, creating parametric relations for campus outdoor objects, and setting up the fitness functions representing design performance objectives. Design optimization is an iterative computational process, using the modelling and simulation results of campus users as performance metrics to generate optimal design solutions.

The campus design, modelling, and optimization framework is also an iterative process, with the optimization results informing the updates of modelling and the potential changes of design requirements or performance objectives. Figure 1 shows

the proposed framework for campus design optimization, consisting of decision-making on the design requirements and performance objectives, design parameter identification and analysis, campus environment modelling, campus user modelling, and design optimization.

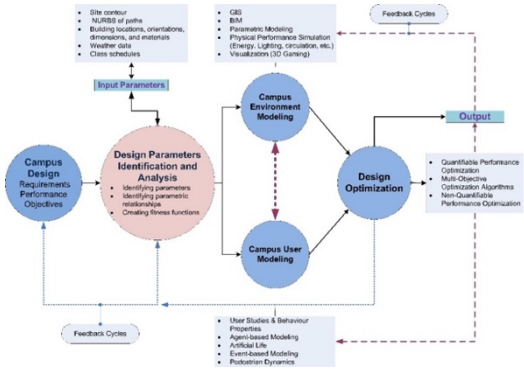

Figure 1. Campus Design Optimization Framework.

3.7. FUNCTIONALITIES AND APPLICATIONS

The functionality of CampuSIM is twofold: a descriptive and predictive tool, and for theory testing and development. It involves a wide range of normal and emergency scenarios of complex, dynamic pedestrian behaviour as follows:

- **Flows:** flow management, evacuation, and traffic path choice/ optimization.
- **Social simulation:** simulation of behaviour on various campus districts.
- **Design:**

1. Improving design elements of pedestrian routes and campus environments: Navigation and moving along a variety of paths across campus zones and across campus/neighbouring zones; managing intersections to identify rules of pedestrian behaviour and characteristic patterns of walking behaviour.
2. Assessment of the impact of new buildings and attractive landscape design enhancements on the overall pedestrian flow.
3. Improved design solutions for entrances, intersections, exits, and egress routes of stadia, theatres, and lecture halls.

4. Conclusion

The significance of the proposed model results from its potential application in a wide range of complex, dynamic pedestrian behaviour scenarios such as flows, social simulations, and design. Its use of multi-objective optimization methods helps fulfil both quantifiable and non-quantifiable design objectives that campus planning and construction need to achieve. In particular, it could be instrumental in addressing the health emergency that resulted from Covid-19 through examining the quality of life, health, relationships, and interactions in public campus spaces. The model would help address many computational challenges, in particular, how to speed up the analysis

cycle to enable each design to be quantitatively analysed coordinated with the design process, which is increasingly accelerated through improved design tools.

The combination of GIS and activity mapping provides a powerful tool supporting designers with empirical evidence of the relationship between environmental design and the use of open space that is spatially explicit and presented in a spatial and visual language familiar to designers. This makes it more likely to be useful and useable.

The proposed model will support the decision-making process as follows:

1. Examining the impact of the environmental qualities on the pedestrians' behaviour and movement. That is essential to better understand how behaviour may vary between environments and to assess the extent to what any differences in behaviour reflect differences in the overall structure of the environment or the characteristics of the people that frequent them.
2. Providing an integrated platform that brings together the design and analysis cycles to produce design insight, Hence, it facilitates effective collaboration among multiple stakeholders to visualize, assess, and communicate the impact of design scenarios quantitatively.
3. Running a large number of scenarios to study the dynamics of the campus zones, test the effectiveness of various design decisions, plans, and policies, and track users' perception and satisfaction along the evaluation period.
4. Pursuing experimental studies for pedestrians' behaviour and dynamic interactions, including theory/data, parameterization, modelling, analysis, validation, and visualization. This is essential to help universities, city planners, and researchers design interventions that reduce barriers and encourage physical activity.
5. Contributing to the identification of the design parameters that can be used in design optimization fitness functions. It will also contribute to constructing the actual fitness functions, considering both quantifiable and non-quantifiable design objectives, for simulating and visualizing the campus activities in exterior spaces.

The planned future work includes prototyping the proposed modelling, simulation, and optimization system and experimenting and evaluating the prototypes with campus planning and design projects. It is critical to emphasize that although the proposed CampuSIM was discussed for campus design, its potential extend beyond that single typology to include a broad range of applications.The advancement of computing software and hardware, including cloud-based, parallel computing, abundant completed CAD-based, and parametric BIM-based optimization studies on smaller design and construction scales makes it feasible to conduct optimization studies on larger scale, more complex master planning and design projects.

References

Abu-Ghazzeh, T. (1999) Communicating Behavioral Research to Campus Design Factors Affecting the Perception and Use of Outdoor Spaces at the University of Jordan. *Environment and Behavior* 31(6):764-804.

Adkins, A., Dill, J., Luhr, G., & Neal, M. (2012) Unpacking walkability: Testing the influence of urban design features on perceptions of walking environment attractiveness. *Journal of Urban Design* 17(4): 499-510. https://doi.org/10.1080/13574809.2012.706365

Alnusairat, S., Al-Shatnawi, Z., & Kakani, A. (2020) Towards an Integrated Sustainable University Campus: The Social Pillar–Case of Al-Ahliyya Amman University Jordan. In Proceedings of the 35th PLEA International Conference: Planning Post Carbon Cities, Online, Coruna, Spain, 1-3 September 2020; Volume 3, pp. 1513–1518.

Birch, D., Kelly, P.H., Field, A.J., & Simondetti, A., (2013). Computationally unifying urban masterplanning. *Proceedings of the ACM International Conference on Computing Frontiers*. ACM, p. 32. https://doi.org/10.1145/2517998

Bonabeau, E. (2002) Agent-Based Modelling: Methods and Techniques for Simulating Human Systems. *Proceedings of the National Academy of Sciences of the United States of America, 99*(10): 7280-7287.

Bruckner, M., Taubock, S., Popper, N., Emrich, S., Rozsenich, B., & Alkilani, S. (2012) Modeling and Simulation of Student Pedestrian Traffic at University Campus. *Simulation Notes Europe* (SNE) 22(2)-B.2012

Deasy, C. (1974) *Design for human affairs*. New York: John Wiley.

Deb, K., Agrawal, S., Pratap, A., & Meyarivan, T. (2000). A fast elitist non-dominated sorting genetic algorithm for multi-objective optimization: NSGA-II. *In Parallel Problem Solving from Nature PPSN VI: 6th International Conference* Paris, France, September 18–20, 2000 Proceedings 6 (pp. 849-858). Springer Berlin Heidelberg.

Duvon, J. (1966 May) The campus landscape. In *American education*. Washington, DC: Department of Health, Education, and Welfare, Education Office.

Eastman, C., Teicholz, P., Sacks, R., & Liston, K. (2011) *BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers, and Contractors.* Hoboken, N.J.: John Wiley and Sons.

Golicnik, B., & Ward Thompson, C. (2010). Emerging relationships between design and use of urban park spaces. *Landscape and Urban Planning*: 94(38-53). https://doi.org/10.1016/j.landurbplan.2009.07.016

Hajrasouliha, A. (2017) Campus score: Measuring university campus qualities. *Landscape Urban Planning*. 158: 166-176. https://doi.org/10.1016/j.landurbplan.2016.10.007

Holzer, D. (2015) BIM and Parametric Design in Academia and Practice: The Changing Context of Knowledge Acquisition and Application in the Digital Age. *International Journal of Architectural Computing* 13(1):65-82.

Jalalian, A., Chalup, S., & Ostwald, M. (2011) Architectural evaluation of simulated pedestrian spatial behavior. *Architectural Science Review* 54(2):132-140 https://doi.org/10.1080/00038628.2011.582372

Kim, J. I., Kim, J., Fischer, M., & Orr, R. (2015) BIM-based decision-support method for master planning of sustainable large-scale developments. *Automation in Construction*, *58*: 95-108. https://doi.org/10.1016/j.autcon.2015.07.003

Koltsova, A., Kunze, A., & Schmitt, G. (2012a) Design of Urban Space at Pedestrian Scale: A Method for Parameterization of Urban Qualities. *Information Visualization, Proceedings of IV2012 16th International Conference*. Montpelier, France: 403-409.

Koltsova, A., Tuncer, B., Georgakopoulou, S., & Schmitt, G. (2012b) Parametric tools for conceptual design support at the pedestrian urban scale. *Digital Aids to Design Creativity*, *Proceedings of the 30th eCAADe 2012*: 279-287.

Manning, K., & Coleman-Boatwright, P. (1991) Student affairs initiatives toward a multicultural university. *Journal of College Student Development* 32(4):367-374.

Meagher, M. (2015) Open Design: Thoughts on software and the representation of movement. In P. Jones and M. Meagher (Eds.) *Architecture and Movement: The Dynamic Experience of Buildings and Landscapes*: 258-265, Routledge, Oxon.

Oppio, A., Forestiero, L., Sciacchitano, L., & Dell'ovo, M. (2021) How to assess urban quality: a spatial multicriteria decision analysis approach, *Valori e Valutazioni*, 28: 21-30

Osaragi, T. (2004, July) Modeling of pedestrian behavior and its applications to spatial evaluation. In *Proceedings of the Third International Joint Conference on Autonomous Agents and Multiagent Systems* 2:836-843. IEEE Computer Society.

Özkan, D.G., Alpak, E.M., &Var, M. (2017) Design and construction process in campus open spaces: A case study of Karadeniz Technical University. *Urban Design International*, 22: 236-252. https://doi.org/10.1057/s41289-017-0041-0

Radford, A., & Gero, J. (1987) *Design by Optimization in Architecture, Building, and Construction*. New York: Wiley.

Ronald, N., Sterling, L., & Kirley, M. (2007) An agent-based approach to modeling pedestrian behavior. *International Journal of Simulation* 8(1):25-38.

Ronald, N. (2007) *Agent-based approaches to pedestrian modeling*, Department of Computer Science and Software Engineering, The University of Melbourne, MS Engineering.

Speake, J., Edmondson, S., & Nawaz, H. (2013) Everyday Encounters with Nature: Students' Perceptions and Use of University Campus Green Spaces. *Human Geographies*7(1):21-31 https://doi.org/10.5719/HGEO.2013.71.21

Speranza, P. (2015) Using parametric methods to understand place in urban design courses. *Journal of Urban Design*, *21*(5), 661-689. https://doi.org/10.1080/13574809.2015.1092378

Steinø, N., & Obeling, E. (2014) Developing a Parametric Urban Design Tool Developing a Parametric Urban Design Tool. *Architecturae Et Artibus* 2014(1): 50-57

Su, Z., &Yan, W. (2015) A fast genetic algorithm for solving architectural design optimization problems. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing* 29(04):457-469. http://dx.doi.org/10.1017/S089006041500044X

Timms, P. & Tight, M. (2010) Aesthetic aspects of walking and cycling, *Built Environment*, 36(4):487-503.

Yan, W. & Jeong, W. (2016) Streamlining Parametric Modeling, BIM, and Building Energy Modeling. *Analytics for Building-Scale Sustainable Ecosystems*, Begell House, New York. Connecticut. USA. https://doi.org/10.13140/RG.2.2.28295.91046

Yan W, Culp C, Graf R (2011) Integrating BIM and Gaming for Real-Time Interactive Architectural Visualization, In *Automation in Construction* 20(4):446-458.Yan W, Kalay Y (2005) Simulating Human Behavior in Built Environments. In *Proceedings of Computer-Aided Architectural Design Futures (CAAD Futures 2005)* Vienna, Austria. pp. 301-310.

Yan, W. (2008) Environment-Behavior Simulation: from CAD to BIM and Beyond. *Proceedings of the Association for Computer Aided Design in Architecture (ACADIA) 2008*, Minneapolis, MN. 478-487.

Yan, W. (2014) Parametric BIM SIM: integrating parametric modeling, BIM, and simulation for architectural design." In *Building Information Modeling: BIM in Current and Future Practice,* Wiley, Hoboken, New Jersey: 57-77. https://doi.org/10.1002/9781119174752.CH5

Yan, W., Rahmani, M., Su, Z, & Altabtabai, J. (2015) Towards Multi-Objective Optimization for Sustainable Buildings with both Quantifiable and Non-Quantifiable Design Objectives. *First International Symposium on Sustainable Human-Building Ecosystems, Pittsburgh, PA, National Science Foundation (NSF) Research Coordination Network (RCN) on Sustainable Human-Building Ecosystems*, Carnegie Mellon Univ., October 5-6.

Zacharias, J. (2001) Pedestrian behavior pedestrian behavior and perception in urban walking environments. *Journal of Planning Literature*, 16(1):3-18.

Zhou, Y. (2008) Agent-based modeling and simulation for pedestrian movement behaviors in Space: a review of applications and GIS issues. In *Proceedings. SPIE* 7143, *Geoinformatics 2008 and Joint Conference on GIS and Built Environment: Geo-Simulation and Virtual GIS Environments*: 71431I-71431I. https://doi.org/10.1117/12.812583